CERTUS
FLEX LIQUID DISPENSER

User Manual

CERTUS RIVO

CERTUS RIVO

12.0 | March 2026

© Fritz Gyger AG. All rights reserved.

Table of Contents

1. 1. Introduction 4
2. 2. Table of Figures 6
3. Products 11
3.1 Certus Rivo 11
3.1.1 Transportation and Storage 0
3.1.2 Technical Specifications 0
3.1.3 Assembly and Operation 0
3.1.4 Certus Control 0
3.1.5 Operation 0
3.1.6 Troubleshooting 0
3.1.7 Maintenance 0
3.1.8 Decommissioning 0
3.1.9 Start-Up 0
3.1.10 Appendix 0

1. 1. Introduction

You have decided for a reliable and high-quality dispenser with highly precise Gyger micro valve technology.

CERTUS RIVO is a fully automatic dispenser for contactless dispensing of various fluids into well plates. Our electromagnetically controlled micro valves enable you to precisely dispense up to eight fluids from 50nl. Automatic dispensing is controlled by **CERTUS CONTROL** software. The unit has been developed to support you with serial dispensing of very small amounts under laboratory conditions.

This knowledgebase will help you optimize the use of **CERTUS RIVO** and avoid mistakes. Most questions are answered in this user manual.

If you have any question or need support, please contact your distributor.

2. 2. Table of Figures

2. 2. Table of Figures

Figure	Category	Caption
Figure 1	Image	Safety Instructions Template
Figure 2	Image	CERTUS CONTROL, Log-In Window
Figure 3	Image	Library Manager, Fluid
Figure 4	Image	Library Manager, Input Area
Figure 5	Image	Library Manager, Micro Valves
Figure 6	Image	Library Manager, Input Area
Figure 7	Image	Library Manager, Well Plates
Figure 8	Image	Well Plates, Standard SBS
Figure 9	Image	Well Plates, Offsets and Pitches
Figure 10	Image	Well Plates Settings
Figure 11	Image	Library Manager, Editing the Evaporation Curtain
Figure 12	Image	Library Manager, Calibration
Figure 13	Image	Library Manager, Calibrations Filtering Possibilities
Figure 14	Image	Library Manager, Calibrations Input Area
Figure 15	Image	Settings, Default Values
Figure 16	Image	Settings, Calibration
Figure 17	Image	Calibration, Valve Display
Figure 18	Image	Settings, Protocol Editor Settings
Figure 19	Image	Settings, Maintenance Settings
Figure 20	Image	Settings, Flush Settings
Figure 21	Image	Settings, Log Settings
Figure 22	Image	Settings, Report Settings
Figure 23	Image	Settings, Device Settings
Figure 24	Image	Settings, Information Settings
Figure 25	Image	Settings, IP Address Settings
Figure 26	Image	Settings, Well Plate Holder
Figure 27	Image	Help
Figure 28	Image	Users
Figure 29	Image	Cirtus Control Login Window
Figure 30	Image	CERTUS CONTROL Logon Window
Figure 31	Image	Selecting a User or Group

Figure	Category	Caption
Figure 32	Image	Selecting a User or Group, Extension Window
Figure 33	Image	Protocol Editor, Menu Bar
Figure 34	Image	Protocol Editor, Status Bar
Figure 35	Image	Operating Principle
Figure 36	Image	Protocol Editor Configuration
Figure 37	Image	CERTUS CONTROL Template Manager
Figure 38	Image	Protocol Editor, CERTUS Calibrator
Figure 39	Image	Protocol Editor, CERTUS Calibrator Configuration
Figure 40	Image	Protocol Editor, CERTUS Calibrator Configuration
Figure 41	Image	CERTUS Calibrator, Flowrate-Graph
Figure 42	Image	CERTUS Calibrator, Volume-Graph
Figure 43	Image	CERTUS Calibrator, Metadata
Figure 44	Image	CERTUS Calibrator, Calibration Point Volume Edit
Figure 45	Image	CERTUS Calibrator, Custom Calibration by Volume
Figure 46	Image	CERTUS Calibrator, Calibration Selector
Figure 47	Image	Configuration Manager - Channels Menu
Figure 48	Image	Protocol Editor, Configuration Manager
Figure 49	Image	Configuration Manager, Fluid Supply Kits
Figure 50	Image	Configuration Manager, Channel Configuration
Figure 51	Image	Calibration Manager, Select Current Calibration
Figure 52	Image	Maintenance Flyout
Figure 53	Image	Maintenance, Definition
Figure 54	Image	Maintenance, Preparation and Cleaning
Figure 55	Image	Maintenance Air
Figure 56	Image	Maintenance, Live Log Flyout
Figure 57	Image	Protocol Editor, Summary View
Figure 58	Image	Interval Purge Flyout
Figure 59	Image	Interval Purge, Head Behavior
Figure 60	Image	Protocol Editor, Protocol
Figure 61	Image	Protocol Editor, Definitions
Figure 62	Image	Protocol Context Menu

2. 2. Table of Figures

Figure	Category	Caption
Figure 63	Image	Protocol, Region Editor
Figure 64	Image	Add Dispense Step
Figure 65	Image	Region Editor, Synchronous Dispense
Figure 66	Image	Dispensing using Fluid by Fluid option
Figure 67	Image	Dispensing using Mixed Fluid option
Figure 68	Image	Regional Editor, Axial Movement
Figure 69	Image	Region Editor, Sequence Replicate
Figure 70	Image	Linear Increase in Dispensing Volume
Figure 71	Image	Region Editor, Discrete Volumes Editor
Figure 72	Image	Interfaces - Cables and Tubes
Figure 73	Image	Overview - How to create an Experiment
Figure 74	Image	Library Manager, Fluid
Figure 75	Image	Library Manager, Add Fluid
Figure 76	Image	Library Manager, Editing Fluid
Figure 77	Image	Library Manager, Delete Fluid
Figure 78	Image	Library Manager, Add Temporary Fluid
Figure 79	Image	Library Manager, Valve
Figure 80	Image	Library Manager, Add New Micro Valve
Figure 81	Image	Library Manager, Delete Micro Valve
Figure 82	Image	Library Manager, Add Temporary Micro Valve
Figure 83	Image	Library Manager, Well Plate
Figure 84	Image	Library Manager, Add New Well Plate
Figure 85	Image	Adding Evaporation Curtain
Figure 86	Image	Editing Evaporation Curtain
Figure 87	Image	Evaporation Curtain, Outer Moat
Figure 88	Image	Adding New Dispensing Point 1
Figure 89	Image	Dispensing Pattern, Vertical
Figure 90	Image	Dispensing Points 1 and 2 Added
Figure 91	Image	Adding New Dispensing Point 3 and 4
Figure 92	Image	Evaporation Curtain, Inter Wells
Figure 93	Image	Library Manager, Calibrations

2. 2. Table of Figures

Figure	Category	Caption
Figure 94	Image	Dispensing Head Configurations
Figure 95	Image	Template Manager, Save Active Configuration
Figure 96	Image	Overview of how to create a Calibration
Figure 97	Image	Calibration Name Adjustment
Figure 98	Image	CERTUS Calibrator Window
Figure 99	Image	Calibration Metadata
Figure 100	Image	Flowrate Graph
Figure 101	Image	Volume Graph
Figure 102	Image	WEB CONTROL, Control Page
Figure 103	Image	Web Control Channels and Preparation
Figure 104	Image	Web Control Cleaning, Air, and Axis
Figure 105	Image	WEB CONTROL, error message (example)

3. Products

3.1 Certus Rivo

3.1.1 ABOUT THIS DOCUMENTATION

This section provides general information on the use of this user manual.

It also contains important information on the documentation itself, such as the presentation of information, the safety standards and directives applied, and the copyright.

Draft

Additional Documents needed

In addition, the following documents are needed to ensure the safe use of the unit:

- Micro valve datasheet, document identification: *CERTUSAN002SMLD300GCDEVxxx.pdf*
- CERTUS RIVO product catalog: *CERTUS RIVO ProductCatalog EN Vxxx.pdf*

General

Language version

The original user manual of the unit is written in German. All other language versions are translations of the original user manual!

In case of doubt, the information contained in the original user manual should be prevailing.

Electronic documentation

The original user manual is available in printout form and electronically in PDF format.

Copyright

All rights reserved. This user manual must not be reproduced or converted, either entirely or in part, into electronic or machine-readable form without written approval from FritzGyger AG.

Responsibility for the documentation

Fritz Gyger AG	Tel: +41 (0)33 336 22 77
Bodmerstrasse 12	info@fgyger.ch
3645 Gwatt (Thun)	www.fgyger.ch
Switzerland	

Maintenance and storage of documentation

To make sure that the user manual is at any time complete and up to date, please do not remove any of its documents. Any new additional documents should be immediately filed. Keep the user manual in an easily accessible place near the unit.

Maintenance

The operational availability and value retention of the unit largely depend on diligent and competentmaintenance.

In addition to its maintenance and care,the unit has to be properly operated if operating personnel and other persons are to remain safe.